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PAPER_1014: SMBH Merger F_U_Bi — Inspiral, Coalescence, Ringdown

Abstract

We compute the UQFF buoyancy force F_{U_Bi} across all three phases of a supermassive black hole (SMBH) binary merger ($M_1 = M_2 = 3.5 \times 10^7 M_{\text{sun}}$). The inspiral phase yields $F_{\text{total}} = 6.98 \times 10^{20} \text{ N}$ with buoyancy damping factor 0.333 and accumulated phase lag 367.0 cycles. During coalescence, buoyancy-induced mass ejection $\Delta M_{\text{buoy}} = 4.05 \times 10^4 \text{ kg}$ is computed. The ringdown phase shows a quasi-normal mode frequency $f_{\text{QNM}} = 2.19 \times 10^{-4} \text{ Hz}$ with SCm correction $\Delta f/f = 9.03 \times 10^{-3}$.

1. Inspiral Phase

The buoyancy force during orbital decay follows:

$$F_{U_Bi}(r) = G * M_1 * M_2 / r^2 * (1 + \text{BETA}_I * S26_3) * D(\text{Gamma})$$

where $D(\text{Gamma})$ is the frequency-dependent damping factor. At peak coupling, $D = 0.333$ indicating 66.7% buoyancy suppression of GW emission.

2. Coalescence Phase

During merger, the buoyancy field ejects mass:

$$\Delta M_{\text{buoy}} = \text{BETA}_I * M_{\text{total}} * (v_{\text{kick}} / c)^2 * S26_3$$

yielding $M_{\text{remnant}} = 6.76 \times 10^7 M_{\text{sun}}$ (3.4% mass deficit from GW + buoyancy radiation).

3. Ringdown Phase

The QNM frequency receives an SCm correction:

$$f_{\text{QNM}} = f_{\text{Kerr}} * (1 + \text{SCm} * \text{BETA}_I * S26_3 / (1 + a^2))$$

with $\Delta f/f = 9.03 \times 10^{-3}$, potentially detectable by LISA.

4. Implementation

File: `fubi_smbh_mergers.py`, class `SMBHInspiralFUBiCalc` (inspiral), `SMBHCoalescenceFUBiCalc` (coalescence), `SMBHRingdownFUBiCalc` (ringdown). CP4 class #598. Tests: 8/8 pass.